

- Document Control
- System Name: Application
- Technical Requirements

TR-SDL-001

- Related Functional Requirement: FR-SDL-001
- Objective: Read source CSV data and load to delta tables (customer and order)
- Target Cluster Configuration:
 - Cluster Name:
 - Runtime Version:
 - Node Type:
 - Driver Node:
 - Worker Nodes:
 - Autoscaling:
 - Auto Termination:
 - Libraries:
- Source Data Details:
 - customer: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata
 - order: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
- Target Data Details:
 - Delta tables: customer, order
- Job Flow:
 1. Read customer CSV from source path
 2. Read order CSV from source path
 3. Load customer data to delta table
 4. Load order data to delta table
- Transformations & Logic:
 - Apply CSV schema as defined below
 - customer: CustId, Name, EmailId, Region
 - order: OrderId, ItemName, PricePerUnit, Qty, Date, CustId
- Error Handling & Logging:

- Log failed loads
- Report schema mismatches
- Log missing files

TR-SDL-002

- Related Functional Requirement: FR-SDL-002
- Objective: Add "TotalAmount" column in "order" table ($\text{PricePerUnit} * \text{Qty}$)
- Target Cluster Configuration:
 - Cluster Name:
 - Runtime Version:
 - Node Type:
 - Driver Node:
 - Worker Nodes:
 - Autoscaling:
 - Auto Termination:
 - Libraries:
- Source Data Details:
 - order delta table
- Target Data Details:
 - order delta table (with TotalAmount column)
- Job Flow:

1. Calculate TotalAmount for each record in the order table

- Transformations & Logic:
 - $\text{TotalAmount} = \text{PricePerUnit} * \text{Qty}$
- Error Handling & Logging:
 - Log calculation errors
 - Handle nulls for PricePerUnit and Qty

TR-SDL-003

- Related Functional Requirement: FR-SDL-003
- Objective: Remove Null and Duplicate records from both tables
- Target Cluster Configuration:

- Cluster Name:
- Runtime Version:
- Node Type:
- Driver Node:
- Worker Nodes:
- Autoscaling:
- Auto Termination:
- Libraries:
- Source Data Details:
- customer delta table
- order delta table
- Target Data Details:
- Cleaned customer and order delta tables

- Job Flow:

1. Identify records with null fields

2. Remove null records

3. Identify duplicates

4. Remove duplicate records

- Transformations & Logic:
- Null record detection
- Duplicate record detection logic
- Error Handling & Logging:
- Log records removed for nulls and duplicates

TR-SDL-004

- Related Functional Requirement: FR-SDL-004
- Objective: Create the "ordersummary" table if not exists
- Target Cluster Configuration:
- Cluster Name:
- Runtime Version:
- Node Type:

- Driver Node:
- Worker Nodes:
- Autoscaling:
- Auto Termination:
- Libraries:
- Source Data Details:
- Target Data Details:
- catalog="gen_ai_poc_databrickscoe"
- schema="sdlc_wizard"
- table=ordersummary
- Job Flow:

1. Check if ordersummary exists

2. Create ordersummary with schema CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date

- Transformations & Logic:
- Error Handling & Logging:
- Log table creation events

TR-SDL-005

- Related Functional Requirement: FR-SDL-005
- Objective: Join customer and order data using CustId and load to SCD type 2 table ordersummary
- Target Cluster Configuration:
- Cluster Name:
- Runtime Version:
- Node Type:
- Driver Node:
- Worker Nodes:
- Autoscaling:
- Auto Termination:
- Libraries:
- Source Data Details:
- customer delta table

- order delta table
- Target Data Details:
- ordersummary table
- Job Flow:

1. Join customer and order on CustId

2. Load joined data to ordersummary (SCD type 2 logic)

- Transformations & Logic:
- SCD type 2 handling
- Join customer and order entities
- Error Handling & Logging:
- Log join failures
- Log history maintenance activity

TR-SDL-006

- Related Functional Requirement: FR-SDL-006
- Objective: Update ordersummary (SCD2) whenever customer table changes
- Target Cluster Configuration:
- Cluster Name:
- Runtime Version:
- Node Type:
- Driver Node:
- Worker Nodes:
- Autoscaling:
- Auto Termination:
- Libraries:
- Source Data Details:
- customer table
- ordersummary table
- Target Data Details:
- Updated ordersummary table
- Job Flow:

1. Monitor changes in customer data
2. Mark old records as Inactive (set EndDate)
3. Insert new records as Active (set StartDate)
4. Maintain full history of changes
 - Transformations & Logic:
 - SCD type 2 logic for Active/Inactive status
 - Error Handling & Logging:
 - Log updates and status changes

TR-SDL-007

- Related Functional Requirement: FR-SDL-007
- Objective: Create "customeraggregatespend" table if not exists
- Target Cluster Configuration:
 - Cluster Name:
 - Runtime Version:
 - Node Type:
 - Driver Node:
 - Worker Nodes:
 - Autoscaling:
 - Auto Termination:
 - Libraries:
- Source Data Details:
- Target Data Details:
 - catalog="gen_ai_poc_databrickscoe"
 - schema="sdlc_wizard"
 - table=customeraggregatespend
- Job Flow:
 1. Check if customeraggregatespend exists
 2. Create with columns: Name, TotalAmount, Date
- Transformations & Logic:

- Error Handling & Logging:
- Log table creation events

TR-SDL-008

- Related Functional Requirement: FR-SDL-008
- Objective: Aggregate TotalAmount from ordersummary, group by Name and Date, load to customeraggregatespend
- Target Cluster Configuration:
 - Cluster Name:
 - Runtime Version:
 - Node Type:
 - Driver Node:
 - Worker Nodes:
 - Autoscaling:
 - Auto Termination:
 - Libraries:
- Source Data Details:
 - ordersummary table
- Target Data Details:
 - customeraggregatespend table
- Job Flow:

1. Aggregate TotalAmount by Name and Date from ordersummary

2. Load results to customeraggregatespend

- Transformations & Logic:
 - Group by Name, Date; sum TotalAmount
- Error Handling & Logging:
 - Log aggregation steps
 - Log load success/failure

TR-SDL-009

- Related Functional Requirement: FR-SDL-009
- Objective: Implement the above processes using Delta Live Tables

- Target Cluster Configuration:

- Cluster Name:
- Runtime Version:
- Node Type:
- Driver Node:
- Worker Nodes:
- Autoscaling:
- Auto Termination:
- Libraries:

- Source Data Details:

- All above tables

- Target Data Details:

- Live tables for all flows

- Job Flow:

1. Apply delta live tables to all ETL steps above

- Transformations & Logic:

- Delta live table orchestration

- Error Handling & Logging:

- Log orchestration events
- Capture and report errors at each step